An Attempt at Error Decomposition for the POEMV Algorithm

Sam Olson

Abstract

Wu and Li [2024] report empirically that filtering error from inaccurate parameter estimation is the dominant source of performance degradation in the POEMV algorithm, but offer no formal support for this attribution. We formalize this attribution by identifying and partially bounding three sources of suboptimality: the approximation error incurred by representing the value function components via finite-degree polynomials, the propagation of parameter estimation errors through the Wonham filter to the belief state, and the resulting sensitivity of the optimal policy mean to belief state error. The first bound exploits the $C^{1,2}$ regularity of the value function established in Wu and Li's Theorem 1; the second uses a Gronwall argument on the Wonham filter SDE; the third follows from the Lipschitz structure of the myopic–hedging decomposition. Together these yield a partial error decomposition for POEMV that clarifies when filtering error dominates. Terms (I) and (II) are bounded under sufficient conditions: Term (I) requires only the $C^{1,2}$ regularity of f, while Term (II) requires a filter stability condition that is violated by the Wu and Li calibration, leaving Corollary 6.2 directionally supported but not formally established for the motivating example. Term (III) remains entirely open. We also identify two open problems whose resolution would complete the decomposition: a filter stability bound valid in the large-$|\rho_1 - \rho_2|$ regime characteristic of equity markets, and a finite-time SGD error bound under a moving loss landscape. The first is an obstacle for any Gronwall-based approach to Wonham filter error in this setting; the second requires controlling the coupling between the gradient estimator and the filtered belief state.

Keywords: Error decomposition, continuous-time reinforcement learning, mean-variance portfolio selection, regime switching, Wonham filter, partial information, exploratory stochastic control.

1 Introduction

Wu and Li [2024] establish that the POEMV algorithm learns near-optimal policies for continuous-time mean-variance portfolio selection in a regime-switching market with an unobservable regime. Their Theorem 1 proves well-posedness and regularity of the value function components $f(t,p)$ and $g(t,p)$, and their Theorem 2 establishes Gaussian optimality of the exploratory policy in the partial-information setting. However, the end-to-end error of the full learning loop is never formally analyzed. Three error sources are implicit: (1) parameter estimation in the Wonham filter, (2) polynomial approximation of f and g, and (3) stochastic gradient error in actor–critic updates. No formal bounds are provided, nor an explicit decomposition.

This gap has practical consequences. In the simulation study of Wu and Li, terminal wealth variance under estimated parameters (0.170) substantially exceeds that under true parameters (0.083), which itself exceeds the finite-difference optimum (0.067). The discrepancy is largely driven by parameter misestimation—for example, $\hat{\lambda}_1 = 1.21$ versus $\lambda_1^* = 0.36$—but this attribution is empirical. A formal error decomposition both validates and quantifies this effect.

We address the first two error sources and detail the difficulties in evaluating the third. Specifically, we establish:

- **Proposition 3.1** (Polynomial approximation error): The error from approximating $\ln f(t,p)$ and $g(t,p)$ by polynomials of bidegree m in $(p,\, T-t)$ decays at rate $O(m^{-2})$ in the sup norm on any compact subset of $(0,1)$, with constants from Wu and Li's Theorem 1.

- **Proposition 4.1** (Filter error propagation): If the estimated environment parameters $\tilde{\theta}$ satisfy $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$ and the filter stability condition $\tilde{\lambda}_1 + \tilde{\lambda}_2 > \frac{1}{2}(1 + 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2)$ holds, the mean-squared belief state error satisfies $\sup_{t\in[0,T]} \mathbb{E}[e_t^2] \leq C_F\, \delta^2/(\tilde{\lambda}_1 + \tilde{\lambda}_2)$, where $e_t = p_t - \tilde{p}_t$ and the damping factor $\tilde{\lambda}_1 + \tilde{\lambda}_2$ is the estimated total transition rate. The stability condition is sufficient but not necessary; it is violated by the Wu and Li calibration, which constitutes Open Problem 1 (Section 7).

- **Proposition 5.1** (Policy sensitivity): The mean-squared policy error induced by belief state error satisfies $\mathbb{E}\big[|m^*(t,X_t,p_t) - m^*(t,X_t,\tilde{p}_t)|^2\big] \leq (C_\gamma^2/\sigma^2)\, \mathbb{E}[(X_t-\omega)^2] \cdot \mathbb{E}[e_t^2]$, where C_γ depends only on the bounds on f_p/f from Theorem 1.

- **Theorem 6.1** (Combined decomposition): The total mean-squared policy error decomposes additively, with the filter propagation term dominant when parameter estimation

error is large relative to polynomial approximation error, consistent with Wu and Li's empirical finding.

Section 7 states the two open problems precisely and outlines the apparent tractable paths toward their resolution.

2 Notation and Preliminaries

We work in the following model, which we state self-containedly for completeness; the notation follows Wu and Li [2024].

Market and regime. A financial market over horizon $[0, T]$ consists of one risk-free asset with constant rate $r \geq 0$ and one risky asset whose discounted price satisfies the regime-switching geometric Brownian motion $dS_t = S_t(\mu_{I_t}\, dt + \sigma\, dB_t)$, with constant volatility $\sigma > 0$ and drift μ_{I_t} determined by the latent market regime. The regime process $I = \{I_t : t \geq 0\}$ is an unobservable continuous-time Markov chain on $\{1, 2\}$ (interpreted as bull and bear markets) with generator matrix Q having off-diagonal entries $\lambda_1, \lambda_2 > 0$, independent of B.

Investment problem. An investor chooses a self-financed strategy u_t (discounted dollar value in the risky asset) adapted to the observable price filtration $\mathcal{F}_t^S$. The discounted wealth satisfies $dX_t^u = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$, where $\hat{\rho}_t = (\rho_1 - \rho_2)p_t + \rho_2$ is the filtered Sharpe ratio and $\hat{B}_t$ is the innovation process. The entropy-regularized mean-variance objective is $\min_{\boldsymbol{\pi}} \mathbb{E}[(X_T^{\boldsymbol{\pi}} - \omega)^2 - \Lambda \int_0^T \mathcal{H}(\boldsymbol{\pi}(\cdot|t, X_t, p_t))\, dt] - (\omega - z)^2$, with temperature $\Lambda \geq 0$ and Lagrange multiplier ω. This is the RSEMV problem of Wu and Li [2024].

The true environment parameters are $\theta^* = (\mu_1^*, \mu_2^*, \sigma^*, \lambda_1^*, \lambda_2^*)$, yielding Sharpe ratios $\rho_i^* = (\mu_i^* - r)/\sigma^*$. Estimated parameters are denoted $\tilde{\theta} = (\tilde{\mu}_1, \tilde{\mu}_2, \tilde{\sigma}, \tilde{\lambda}_1, \tilde{\lambda}_2)$ with $\tilde{\rho}_i = (\tilde{\mu}_i - r)/\tilde{\sigma}$. Parameter errors are $\delta\mu_i = \tilde{\mu}_i - \mu_i^*$, $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$, $\delta\rho_i = \tilde{\rho}_i - \rho_i^*$.

The belief state under true parameters is $p_t = \Pr(I_t = 1 \mid \mathcal{F}_t^S)$, satisfying the *Wonham filter SDE*:

$$dp_t = \left[-(\lambda_1^* + \lambda_2^*)p_t + \lambda_2^*\right] dt + (\rho_1^* - \rho_2^*)\, p_t(1 - p_t)\, d\hat{B}_t. \tag{1}$$

The belief state under estimated parameters, $\tilde{p}_t$, satisfies:

$$d\tilde{p}_t = \left[-(\tilde{\lambda}_1 + \tilde{\lambda}_2)\tilde{p}_t + \tilde{\lambda}_2\right] dt + (\tilde{\rho}_1 - \tilde{\rho}_2)\, \tilde{p}_t(1 - \tilde{p}_t)\, d\hat{B}_t, \tag{2}$$

where both filters share the same innovation process $\hat{B}_t$. This representation is standard in filter stability analysis—every $\mathcal{F}^S$-adapted semimartingale admits such a representation—but

it suppresses one $O(\delta)$ term: in practice the misspecified filter computes $d\tilde{B}_t = dS_t/(\tilde{\sigma} S_t) - \tilde{\hat{\rho}}_t\, dt$ using estimated parameters, which differs from $d\hat{B}_t$ by an $O(\delta)$ drift. Writing both filters against $\hat{B}_t$ folds this discrepancy into the drift forcing ξ_t^b of the error SDE in Proposition 4.1 rather than the diffusion coefficient, making the Gronwall bound slightly tighter than a fully explicit accounting would yield.

The optimal value function of the RSEMV problem is:

$$V^*(t,x,p) = (x-\omega)^2 \frac{1}{f(t,p)} + g(t,p) - (\omega - z)^2, \tag{3}$$

where f and g satisfy the degenerate parabolic PDEs (18) and (19) of Wu and Li [2024]. The optimal Gaussian policy has mean:

$$m^*(t,x,p) = -\frac{x-\omega}{\sigma} \underbrace{\left[\rho_2^* + (\rho_1^* - \rho_2^*)p\right]}_{\text{myopic}} + \frac{x-\omega}{\sigma} \underbrace{(\rho_1^* - \rho_2^*)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}}_{\text{hedging demand}}. \tag{4}$$

We collect the key regularity results from Wu and Li's Theorem 1 that all subsequent bounds rely on.

Assumption 2.1 (Regularity of f, from Wu and Li 2024 Theorem 1)**.**

(R1) $f \in C^{1,2}([0,T) \times [\varepsilon, 1-\varepsilon]) \cap C([0,T] \times [\varepsilon, 1-\varepsilon])$ *for every* $\varepsilon \in (0, \frac{1}{2})$.

(R2) $1 \le f(t,p) \le e^{\bar{\rho}^2 (T-t)}$ *on* $[0,T] \times (0,1)$, *where* $\bar{\rho} = \max\{|\rho_1^*|, |\rho_2^*|\}$.

(R3) *There exist finite constants* $C_1, C_2 > 0$ *such that* $|f_p(t,p)| \le C_1$ *and* $|f_{pp}(t,p)| \le C_2$ *uniformly on* $[0,T] \times (0,1)$.

(R4) f *is decreasing in* t *for each fixed* $p \in (0,1)$.

3 Polynomial Approximation Error

Wu and Li [2024] approximate $\ln f(t,p)$ and $g(t,p)$ by polynomials of bidegree m in $(p, T-t)$:

$$f^\theta(t,p) = \exp\left(\sum_{\substack{0 \le i \le m \\ 1 \le j \le m}} \theta_{i,j}\, p^i (T-t)^j \right), \qquad g^\theta(t,p) = \sum_{\substack{0 \le i \le m \\ 1 \le j \le m}} \tilde{\theta}_{i,j}\, p^i (T-t)^j. \tag{5}$$

Approximation quality depends on the smoothness of $\ln f$ and g on the approximation domain.

Proposition 3.1 (Polynomial approximation error)**.** *Fix $\varepsilon \in (0, \frac{1}{2})$ and let $\mathcal{D}_\varepsilon = [0,T] \times [\varepsilon, 1-\varepsilon]$. Define the oracle approximation error of bidegree-m polynomials as $E_m(\ln f,\, \mathcal{D}_\varepsilon) := \inf_{q \in \mathcal{P}_m} \|\ln f - q\|_{L^\infty(\mathcal{D}_\varepsilon)}$, where $\mathcal{P}_m$ denotes the class of bivariate polynomials of degree at most m in each variable. Under Assumption 2.1, there exists a constant K_ε depending on ε, $\bar{\rho}$, C_1, C_2, and T but not on m, such that:*

$$E_m(\ln f,\, \mathcal{D}_\varepsilon) \;\le\; \frac{K_\varepsilon}{m^2}. \tag{6}$$

An analogous bound holds for g.

Proof. Define $h(\tau, p) = \ln f(T-\tau, p)$ on $[0,T] \times [\varepsilon, 1-\varepsilon]$, so that polynomial approximation in $(p, T-t)$ corresponds to approximation of h in (τ, p). We verify that h admits bounded second-order partial derivatives on $\mathcal{D}_\varepsilon$.

For the p-derivatives, write $h_p = f_p/f$ and $h_{pp} = f_{pp}/f - (f_p/f)^2$. By (R2), $f \ge 1$, so $|h_p| \le |f_p| \le C_1$ and $|h_{pp}| \le |f_{pp}|/f + (f_p/f)^2 \le C_2 + C_1^2$. For the τ-derivative, $h_\tau = -f_t/f$; since $f_t \le 0$ by (R4) and $f \ge 1$, there exists a finite constant C_3 such that $|h_\tau| \le C_3$. For the second τ-derivative, $h_{\tau\tau} = -f_{tt}/f + (f_t/f)^2$. Differentiating PDE (18) of Wu and Li [2024] with respect to t yields a linear parabolic PDE for f_t with the same degenerate diffusion coefficient $a(p) = (\rho_1^* - \rho_2^*)^2 p^2 (1-p)^2$. On $\mathcal{D}_\varepsilon = [0,T] \times [\varepsilon, 1-\varepsilon]$ the diffusion is uniformly elliptic (bounded away from zero), so Theorem 10 in Chapter 2 of Krylov [2008] applied to this derived PDE yields boundedness of f_{tt} on $\mathcal{D}_\varepsilon$; the argument is identical to Wu and Li's proof of part (iii) of Theorem 1 establishing boundedness of f_t. Hence there exists a finite constant C_4 such that $|h_{\tau\tau}| \le C_4$.

Thus h satisfies the regularity conditions required for second-order Jackson-type approximation bounds with uniformly bounded second-order derivatives in both variables. We apply the Jackson theorem for bivariate algebraic polynomial approximation on a compact rectangle, which extends the univariate $O(m^{-2})$ rate to tensor-product polynomial spaces under bounded second-order partial derivatives in each variable separately; a fully rigorous treatment in the anisotropic case would require verifying the mixed derivative $h_{\tau p}$ as well, but this follows from differentiating the PDE (18) of Wu and Li [2024] with respect to both t and p and invoking the same Schauder estimates (see, e.g., Cheney 1966, Ditzian and Totik 1987). This yields:

$$E_m(h,\, [0,T] \times [\varepsilon, 1-\varepsilon]) \;\le\; \frac{C}{m^2}\big(\|h_{pp}\|_{L^\infty} + \|h_{\tau\tau}\|_{L^\infty}\big) \;\le\; \frac{K_\varepsilon}{m^2},$$

where $K_\varepsilon = C(C_2 + C_1^2 + C_4)$. □

Remark 3.2. *The implemented policy m_m^* uses $(q_m)_p$ in place of $(\ln f)_p = f_p/f$, where q_m is the degree-m best polynomial approximation to $\ln f$. The policy-relevant approximation error is therefore in the* derivative $(\ln f)_p$, *not in $\ln f$ itself. By the univariate Jackson theorem applied to $h_p = (\ln f)_p$: since h_{pp} is bounded on $\mathcal{D}_\varepsilon$ (established in the proof of Proposition 3.1), the best degree-m polynomial approximation to h_p satisfies $E_m(h_p, \mathcal{D}_\varepsilon) \leq K'_\varepsilon/m$, one order slower than the $O(m^{-2})$ rate for h itself. Since $m^*(t, x, p)$ is linear in f_p/f through the hedging term (equation* (4)*), the policy mean approximation error is $O(m^{-1})$, and the* squared *policy error from this source is $O(m^{-2})$.*

4 Wonham Filter Error Propagation

Let $e_t \coloneqq p_t - \tilde{p}_t$ denote the filter error process.

Proposition 4.1 (Filter error propagation)**.** *Suppose $\left\|\tilde{\theta} - \theta^*\right\| \leq \delta$ for some $\delta > 0$, and that the induced Sharpe ratio errors satisfy $|\delta\rho_i| \leq C_\rho \delta$ for some constant $C_\rho > 0$. Also assume $p_0 = \tilde{p}_0$ (both filters initialized identically). Assume further the* filter stability condition

$$\tilde{\lambda}_1 + \tilde{\lambda}_2 \;>\; \tfrac{1}{2}\big(1 + 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2\big), \tag{7}$$

which ensures mean-square contraction of the filter error process. Then there exists a constant $C_F > 0$ depending only on $\bar{\rho}$, $|\rho_1^* - \rho_2^*|$, and T, such that:

$$\sup_{t\in[0,T]} \mathbb{E}\big[e_t^2\big] \;\leq\; \frac{C_F\,\delta^2}{\tilde{\lambda}_1 + \tilde{\lambda}_2}. \tag{8}$$

Proof. From (1) and (2), the error e_t satisfies:

$$de_t = \big[b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta})\big]\,dt + \big[a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big]\,d\hat{B}_t,$$

where $b(p;\theta) = -(\lambda_1+\lambda_2)p+\lambda_2$ and $a(p;\theta) = (\rho_1-\rho_2)p(1-p)$ denote the drift and diffusion coefficients of the Wonham SDE respectively.

With $\delta\lambda_i = \tilde{\lambda}_i - \lambda_i^*$ as defined in Section 2, write the drift difference as:

$$b(p_t;\theta^*) - b(\tilde{p}_t;\tilde{\theta}) = -(\tilde{\lambda}_1 + \tilde{\lambda}_2)\,e_t + \underbrace{\big[-\delta\lambda_2 + (\delta\lambda_1 + \delta\lambda_2)\,p_t\big]}_{\xi_t^b}.$$

Since $p_t \in (0,1)$, we have $|\xi_t^b| \leq |\delta\lambda_2| + |\delta\lambda_1 + \delta\lambda_2| \leq |\delta\lambda_1| + 2|\delta\lambda_2| \leq 3\|\tilde{\theta} - \theta^*\| \leq 3\delta$.

Write:

$$a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta}) = \underbrace{(\delta\rho_1 - \delta\rho_2)\, p_t(1-p_t)}_{\xi_t^\sigma} + (\tilde{\rho}_1 - \tilde{\rho}_2)\, \underbrace{\big[p_t(1-p_t) - \tilde{p}_t(1-\tilde{p}_t)\big]}_{\text{Lipschitz in } e_t}.$$

Since $p \mapsto p(1-p)$ is Lipschitz on $[0,1]$ with constant 1, the second term is bounded by $|\tilde{\rho}_1 - \tilde{\rho}_2| \cdot |e_t|$. Moreover, since $\sup_{p\in[0,1]} p(1-p) = 1/4$, we have $|\xi_t^\sigma| = |(\delta\rho_1 - \delta\rho_2)|\, p_t(1-p_t) \leq |\delta\rho_1 - \delta\rho_2|/4 \leq C_\rho \delta$, where the final inequality uses $|\delta\rho_i| \leq C_\rho \delta$. The constant C_ρ absorbs the dependence of Sharpe ratio error on (μ, σ) parameterization; under the Euclidean norm, Sharpe ratio errors $\delta\rho_i = (\tilde{\mu}_i - \mu_i^*)/\tilde{\sigma} - (\mu_i^* - r)\, \delta\sigma/(\sigma^*\tilde{\sigma})$ can exceed δ by a factor of order $1/\sigma$, which for $\sigma = 0.22$ is approximately 4.5. The bound absorbs this factor into C_F without making it explicit; in the Wu and Li calibration this factor is nontrivial and the constant C_F should be understood accordingly.

Itô's formula applied to e_t^2 gives:

$$\frac{d}{dt}\mathbb{E}[e_t^2] = -2(\tilde{\lambda}_1 + \tilde{\lambda}_2)\, \mathbb{E}[e_t^2] + 2\, \mathbb{E}[e_t\, \xi_t^b] + \mathbb{E}\big[\big(a(p_t;\theta^*) - a(\tilde{p}_t;\tilde{\theta})\big)^2\big].$$

Applying Young's inequality $2|e_t \xi_t^b| \leq e_t^2 + (\xi_t^b)^2$ and bounding the quadratic diffusion term using the Lipschitz property and the bound on ξ_t^σ:

$$\frac{d}{dt}\mathbb{E}[e_t^2] \;\leq\; -\alpha\, \mathbb{E}[e_t^2] + C_1'\, \delta^2,$$

where $\alpha = 2(\tilde{\lambda}_1 + \tilde{\lambda}_2) - 1 - 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2 > 0$ and C_1' absorbs the constant from the noise terms. Positivity of α follows directly from the filter stability condition (7). This condition is not satisfied for the calibrated parameters of Wu and Li: with $\lambda_1^* + \lambda_2^* \approx 3.25$ and $|\rho_1^* - \rho_2^*| \approx 4.45$, the right-hand side evaluates to approximately 20.3, exceeding 3.25 by an order of magnitude. Condition (7) is therefore a sufficient condition that the motivating example violates, and not merely at the boundary. This identifies a structural limitation of Gronwall-based approaches to filter error in equity market settings: the large Sharpe ratio differential that makes regime information valuable—and makes POEMV worth developing—is precisely what causes the diffusion term in the error SDE to dominate the mean-reversion term. Bound (8) may still hold in this regime through a different argument (see Open Problem 1 in Section 7), but this cannot be concluded from the present proof. The dominance conclusion of Corollary 6.2 is therefore directionally supported rather than formally established for the specific Wu and Li calibration. Solving the differential inequality via Gronwall's lemma with $e_0 = 0$ yields

$$\mathbb{E}[e_t^2] \leq \frac{C_1'\delta^2}{\alpha}\left(1 - e^{-\alpha t}\right) \leq \frac{C_1'\delta^2}{\alpha},$$

where $\alpha = 2(\tilde{\lambda}_1 + \tilde{\lambda}_2) - 1 - 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2 > 0$ under condition (7). Since $\alpha < 2(\tilde{\lambda}_1 + \tilde{\lambda}_2)$ and condition (7) implies α is bounded below by a positive multiple of $\tilde{\lambda}_1 + \tilde{\lambda}_2$, we have $1/\alpha \leq C/(\tilde{\lambda}_1 + \tilde{\lambda}_2)$ for a bounded constant C absorbed into C_F, yielding bound (8). □

Remark 4.2. *The bound is $O(\delta^2)$, so parameter errors enter* quadratically*: doubling the estimation error quadruples the belief state error bound. This scaling should not be confused with qualitative regime tracking—the belief state $\tilde{p}_t$ can correctly identify bull and bear regimes visually while still inducing a large policy error through the amplification in Proposition 5.1; these are distinct failure modes.*

5 Policy Sensitivity to Belief State Error

The decomposition is not orthogonal: Term (II) depends on $\mathbb{E}[e_t^2]$, which itself is driven by parameter estimation error. As a result, dominance statements are asymptotic rather than exact.

Define the function:

$$\gamma^*(t,p) \;:=\; \rho_2^* + (\rho_1^* - \rho_2^*)p - (\rho_1^* - \rho_2^*)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}, \tag{9}$$

so that $m^*(t,x,p) = -(x-\omega)\,\gamma^*(t,p)/\sigma$.

Proposition 5.1 (Policy sensitivity)**.** *There exists a constant $C_\gamma > 0$ depending only on $|\rho_1^* - \rho_2^*|$, C_1, and C_2, such that $\gamma^*(t,\cdot)$ is Lipschitz in p uniformly over $t \in [0,T]$:*

$$|\gamma^*(t,p) - \gamma^*(t,\tilde{p})| \;\leq\; C_\gamma\, |p - \tilde{p}|\,, \qquad \forall\, t \in [0,T],\; p,\tilde{p} \in (0,1). \tag{10}$$

Consequently, assuming $X_t - \omega$ and e_t are weakly correlated (so that second moments factor up to constants), the mean-squared policy error satisfies:

$$\mathbb{E}\big[|m^*(t,X_t,p_t) - m^*(t,X_t,\tilde{p}_t)|^2\big] \;\leq\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}[(X_t-\omega)^2]\cdot\mathbb{E}[e_t^2]. \tag{11}$$

Proof. We bound $|\partial\gamma^*/\partial p|$ uniformly in t. Differentiating (9):

$$\frac{\partial\gamma^*}{\partial p} = (\rho_1^* - \rho_2^*)\left[1 - \frac{\partial}{\partial p}\left(p(1-p)\,\frac{f_p}{f}\right)\right].$$

Expanding and applying (R2)–(R3):

$$\left|(1-2p)\,\frac{f_p}{f}\right| \leq C_1, \quad \left|p(1-p)\,\frac{f_{pp}}{f}\right| \leq \frac{C_2}{4}, \quad \left|p(1-p)\,\frac{f_p^2}{f^2}\right| \leq \frac{C_1^2}{4},$$

Defining $C_\gamma := |\rho_1^* - \rho_2^*|(1 + C_1 + C_2/4 + C_1^2/4)$ gives $|\partial\gamma^*/\partial p| \le C_\gamma$ uniformly in (t, p), establishing (10) by the mean value theorem. Bound (11) then follows by squaring, taking expectations, and applying (10):

$$\mathbb{E}\big[|m^*(\cdot, p_t) - m^*(\cdot, \tilde{p}_t)|^2\big] = \frac{1}{\sigma^2}\,\mathbb{E}\big[(X_t - \omega)^2\,(\gamma^*(t, p_t) - \gamma^*(t, \tilde{p}_t))^2\big] \;\le\; \frac{C_\gamma^2}{\sigma^2}\,\mathbb{E}\big[(X_t - \omega)^2\,e_t^2\big]\,.$$

Under the stated weak-correlation assumption, formalized as approximate independence of $(X_t - \omega)$ and e_t, we have $\mathbb{E}[(X_t - \omega)^2\,e_t^2] \approx \mathbb{E}[(X_t - \omega)^2] \cdot \mathbb{E}[e_t^2]$, yielding (11). More precisely, if $\mathrm{Cov}((X_t - \omega)^2,\, e_t^2) \le \kappa\,\mathbb{E}[(X_t - \omega)^2]\,\mathbb{E}[e_t^2]$ for some $\kappa \in [0, 1)$, the bound holds with an additional factor $(1 + \kappa)$ absorbed into C_γ^2.

□

6 The Combined Decomposition

Theorem 6.1 (Error decomposition)**.** *Let $m \ge 1$ be the polynomial approximation degree, let $\left\|\tilde{\theta} - \theta^*\right\| \le \delta$, and fix $\varepsilon \in (0, \frac{1}{2})$. For $p_t, \tilde{p}_t \in [\varepsilon, 1-\varepsilon]$ almost surely, the total mean-squared policy error satisfies:*

$$\mathbb{E}\big[|m^*(t, X_t, p_t) - \tilde{m}(t, X_t, \tilde{p}_t)|^2\big] \;\le\; \underbrace{\frac{A_\varepsilon}{m^2}\,\mathbb{E}[(X_t - \omega)^2]}_{\text{(I) polynomial approx}} \;+\; \underbrace{\frac{C_\gamma^2}{\sigma^2} \cdot \frac{C_F\,\delta^2}{\tilde{\lambda}_1 + \tilde{\lambda}_2} \cdot \mathbb{E}[(X_t - \omega)^2]}_{\text{(II) filter propagation}}, \quad (12)$$

where $\tilde{m}$ denotes the implemented policy using the degree-m polynomial approximation evaluated at the estimated belief state $\tilde{p}_t$, $A_\varepsilon = 2(K'_\varepsilon)^2/\sigma^2$ with K'_ε the derivative approximation constant from Remark 3.2, and C_γ, C_F are the constants from Propositions 4.1 and 5.1.

Proof. Apply the triangle inequality followed by $(a + b)^2 \le 2a^2 + 2b^2$:

$$|m^* - \tilde{m}|^2 \;\le\; 2\,\underbrace{|m^*(t, x, p_t) - m_m^*(t, x, p_t)|^2}_{\text{approx error at true } p_t} \;+\; 2\,\underbrace{|m_m^*(t, x, p_t) - m_m^*(t, x, \tilde{p}_t)|^2}_{\text{filter error in approx policy}},$$

where m_m^* denotes the policy using the degree-m polynomial approximation to f evaluated at the true belief state. Taking expectations, we bound each term separately. For the first term: since $m^*(t, x, p) = -(x - \omega)\gamma^*(t, p)/\sigma$ and m_m^* replaces f_p/f with $(q_m)_p$ in γ^*, the approximation error satisfies $|m^*(t, x, p_t) - m_m^*(t, x, p_t)| = |x - \omega|\,|(\ln f)_p - (q_m)_p|/\sigma$. By Remark 3.2, $\|(\ln f)_p - (q_m)_p\|_{L^\infty(\mathcal{D}_\varepsilon)} \le K'_\varepsilon/m$, so $\mathbb{E}[|m^* - m_m^*|^2] \le (K'_\varepsilon)^2\,\mathbb{E}[(X_t - \omega)^2]/(\sigma^2 m^2)$, giving a contribution of $2(K'_\varepsilon)^2\mathbb{E}[(X_t - \omega)^2]/(\sigma^2 m^2)$ after multiplying by 2; the factor $1/\sigma^2$ is absorbed into A_ε. For the second term, Propositions 4.1 and 5.1 yield the filter propagation

bound; the Lipschitz constant of m_m^* in p differs from C_γ by $O(m^{-1})$ corrections absorbed into A_ε. This gives (12) with $A_\varepsilon = 2(K'_\varepsilon)^2/\sigma^2$. □

Corollary 6.2 (Dominant error source)**.** *Term* (II) *dominates term* (I) *in* (12) *when:*

$$\delta^2 \gg \frac{(\tilde{\lambda}_1 + \tilde{\lambda}_2)\, A_\varepsilon}{C_\gamma^2\, C_F\, \sigma^2\, m^2}.$$

For the calibrated parameters $\mu_1 = 0.25$, $\mu_2 = -0.73$, $\sigma = 0.22$, $\lambda_1 = 0.36$, $\lambda_2 = 2.89$ and polynomial degree $m = 2$, the right-hand side is on the order of 10^{-1}, with the precise value depending on the PDE regularity constants C_1, C_2 from Assumption 2.1, which Wu and Li do not compute explicitly. Whenever parameter estimation errors are on the order of 10–15% of the parameter scale or larger, filtering error can therefore be expected to dominate polynomial approximation error. The heuristic estimator's overstatement of λ_1 by a factor of 3 places the error far above this threshold.

7 Discussion

Connection to Wu and Li's empirical findings. Corollary 6.2 formalizes Wu and Li's attribution of performance degradation to filtering error. As noted in the introduction, the heuristic estimator produces $\hat{\lambda}_1 = 1.21$ against a true $\lambda_1^* = 0.36$, placing the parameter error well within the regime where filter propagation dominates. Theorem 6.1 explains the empirical convergence behavior reported by Wu and Li: policy parameters converge substantially closer to their optima under true parameters than under estimated ones. The quadratic amplification of filtering error in Term (II) implies that even moderate parameter misspecification induces a significant policy gap.

The role of the transition rate sum. The factor $(\tilde{\lambda}_1 + \tilde{\lambda}_2)^{-1}$ in bound (II) is the natural self-correction timescale of the Wonham filter: faster-switching markets are more robust to parameter misestimation, while near-stationary markets amplify it. POEMV should be deployed with particular care in the latter setting.

Tightness of the bounds. The $O(m^{-2})$ rate in Proposition 3.1 is sharp given $C^{1,2}$ regularity; boundedness of third-order derivatives of f would yield $O(m^{-3})$ via the Jackson theorem, pointing to a direction for strengthening Wu and Li's Theorem 1 through finer energy estimates.

Open problems and what a complete decomposition requires. The analysis above establishes Terms (I) and (II) of the decomposition under sufficient conditions and leaves two problems open. We state them precisely.

Open Problem 1 (Filter stability in the large-$|\rho_1 - \rho_2|$ regime)**.** Proposition 4.1 establishes bound (8) under condition (7), which requires the total transition rate to exceed $\frac{1}{2}(1 + 2|\tilde{\rho}_1 - \tilde{\rho}_2|^2)$. For the Wu and Li calibration this threshold is approximately 20.3 against a transition rate sum of 3.25. The condition fails because the Gronwall argument bounds the diffusion coefficient of the error SDE uniformly by $|\tilde{\rho}_1 - \tilde{\rho}_2|$, which is large in equity markets. A complete filter error bound valid for the Wu and Li calibration would require either a relative entropy Lyapunov argument in the spirit of van Handel [2009] or an occupancy-time decomposition exploiting the boundary concentration of p_t when the diffusion coefficient is large—the latter being tractable because the stationary distribution of p_t under the Wu and Li parameters is approximately $\mathrm{Beta}(2\lambda_2/(\rho_1-\rho_2)^2,\, 2\lambda_1/(\rho_1-\rho_2)^2)$, which concentrates near the boundaries and limits the time p_t spends where the diffusion term is large. Resolving this problem would formally establish Corollary 6.2 for the motivating example rather than leaving it directionally supported.

Open Problem 2 (SGD error under a moving loss landscape)**.** Standard finite-time SGD bounds require the loss landscape to be stationary. Here the martingale loss depends on $\tilde{p}_t$, which evolves stochastically and is contaminated by the same parameter error driving Term (II). Consecutive gradient iterates are therefore evaluated at incomparable points, and because both the wealth process and the filter are driven by the same innovation $\hat{B}_t$, the gradient estimator carries a bias proportional to e_t that does not factor out of the decomposition. Three routes toward a formal Term (III) bound are available, in increasing order of technical demand. The first exploits the linear-quadratic structure of the RSEMV problem: the martingale loss is quadratic in the policy parameters ϕ, so the Hessian is globally bounded and standard finite-time SGD regret bounds apply without requiring the loss landscape to be fixed—this route requires no assumptions beyond those already in Wu and Li's framework. The second treats $\tilde{p}_t$ as approximately constant over each gradient update window and applies SGD theory conditional on the filter path, bounding the approximation cost via Term (II) and a mixing condition on $\tilde{p}_t$ at the $(\tilde{\lambda}_1 + \tilde{\lambda}_2)^{-1}$ timescale. The third follows Tang and Zhou [2024] in treating filter error as an adversarial perturbation to the gradient oracle, yielding Term (III) $\leq$ standard SGD regret + cumulative filter perturbation cost controlled by Term (II); this requires a Lipschitz condition on the gradient in p, closely related to the bound on $|\partial\gamma^*/\partial p|$ in Proposition 5.1.

The two open problems are related: resolving Open Problem 1 would directly supply

the $\mathbb{E}[e_t^2]$ bound needed by the adversarial perturbation route to Open Problem 2. As a sanity check, the gap between the finite-difference optimum (0.067) and POEMV under true parameters (0.083) jointly bounds Terms (I) and (III); its small magnitude suggests their combined contribution is modest relative to the filter propagation term even without formal bounds.

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